

# Passive Optical Networks Systems with Co-Propagating Distributed Acoustic Sensing

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**Abstract** We study the impact of distributed acoustic sensing on PONs. We show that wavelength isolation is required, and how Raman scattering can impact the downstream G-PON. We also provide guiding rules to mitigate DAS impact on PONs. ©2026 The Author(s)

## Introduction

Distributed Fiber Optic Sensing (DFOS) is a collection of techniques enabling the possibility to transform the deployed optical fibers networks in vast sensors exploiting technologies based on backscattering sensing or feed-forward sensing such as Distributed Acoustic Sensing (DAS), Distributed Temperature Sensing (DTS), Distributed Temperature & Strain Sensing (DTSS), or State of Polarization (SOP) [1, 2]. Therefore, it offers the possibility to create value from the already deployed optical infrastructure in two ways: it can either 1) improve network operation in locating or preventing outages for example, but it can also 2) create valuable data for third parties such as scientific applications [3], mining induced seismicity [4],... When considering access networks, the density of this network enables a lot of possibilities: road traffic monitoring, detection and location of water pipes leakages, bridge deformation monitoring, ... Sensing the optical fiber in access networks is possible in several ways: one can either use supernumerary "dark" fibers (already available in deployed cables), or using active transmission fibers, used for example for Fiber To The Home (FTTH). In the latter case, Passive Optical Networks (PONs) form the most represented family of technologies for access networks, using Point-to-Multipoint (PtMP) tree topology based on a splitter having typically a 1:64 split ratio (see Fig. 1), introducing important losses which typical sensing technologies cannot overcome [5]. Moreover, the tree topology of PONs introduces an ambiguity in the event location. When inserting the DFOS

stimuli from the Optical Line Termination (OLT) side (for shelter, security, and power supply reasons), exploiting the sensing technologies in PON is then possible on an active link, but is limited to the "trunk" part of Optical Distribution Network (ODN). One way to achieve this injection is to exploit the unused port of an optical 2:2 coupler (see Fig. 1), which is often present to merge a former 64 Optical Network Units (ONUs) capable ODNs in a 128 ONUs capable ODN, in countries such as France. This 2:2 coupler is for example present in more than half the ODNs in France for Orange network and may be used to introduce Optical Time Domain Reflectometer (OTDR). Another way to exploit PON links is to introduce a coexistence element (see "WDM" on Fig. 1), reusing for example specifications including the RF video band (1550-1560 nm) [6]. In both cases, an optical switch can be used to permit the DFOS system to be shared among different ODNs, thus sharing the high cost of the DFOS system. However, the coexistence of sensing technologies and usual telecoms technologies, such as PON for FTTH, may be harmful.

In this paper, we propose to study the interaction of Rayleigh-based pulse detection technologies such as DAS with PONs, and identify their ability to coexist on the same link. We mainly focus on non-linear interaction triggered by the high peak power of the sensing pulse. Several works already studied interactions between sensing and telecoms: in [7], the authors showed that the nonlinear Kerr-induced modulation instability affected neighbors channels in a Dense Wavelength Division Multiplexing (DWDM) configuration for long-haul trans-

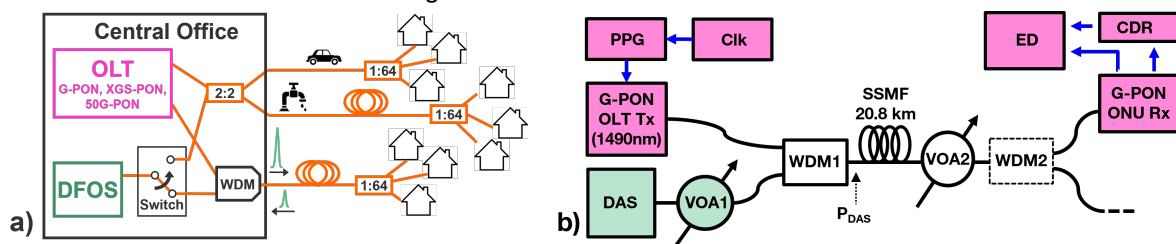
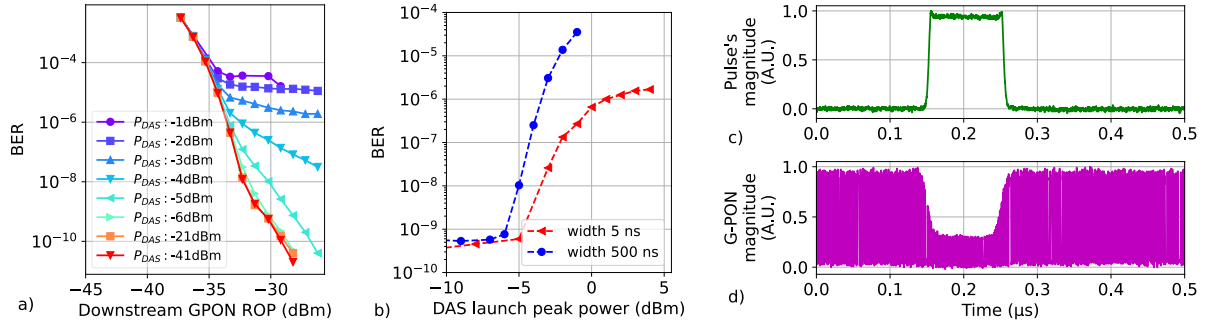


Fig. 1: a) Simplified view of the deployed PONs and DFOS and b) experimental setup.



**Fig. 2:** a) BER vs ROP for various DAS peak power (no WDM filter). b) BER dependence to the pulse width and DAS peak power (ROP=-30.2 dBm). c) DAS pulse and d) corresponding downstream G-PON "notch" after 20.8 km propagation ( $P_{DAS}=27.3$  dBm).

mission. In [8] the authors demonstrated in a field trial that no impact was observed on a 25G-PON system, with a 25G-PON wavelength plan in the O-band. Simultaneous fiber-optic sensing and PON over the entire link using special enhanced scattering fibers, was also demonstrated [9]. Here we mainly focus on Stimulated Raman Scattering (SRS) interactions, considering the full International Telecommunication Union (ITU-T) PON wavelength plan, as we observed that SRS was the main source of interaction in that context.

### Experimental setup

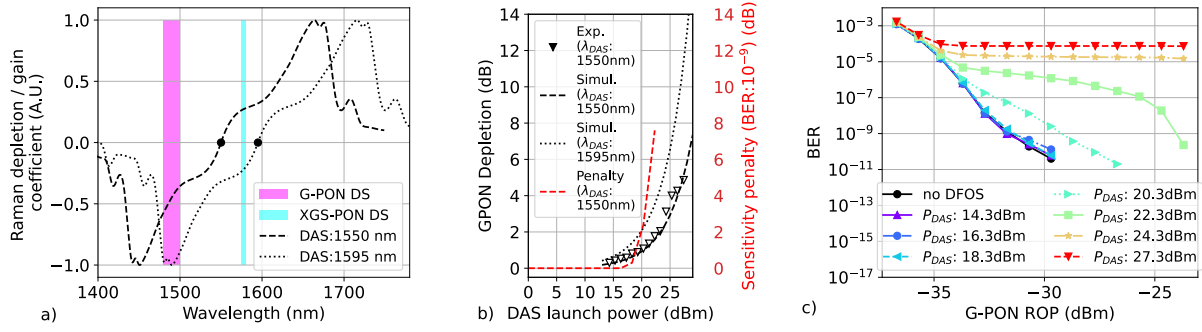
The experimental setup is depicted in Fig. 1.b. It consists on a Gigabit-capable Passive Optical Network (G-PON) link (top, pink) and a DFOS system (bottom, green). The DFOS systems used here are two relatively similar DASs, exploiting either the phase or the amplitude of the backscattered signal. A Pulse Pattern Generator (PPG) at 2.5 Gbit/s with a PRBS31 drives a dual-fiber G-PON transceiver (OLT Tx) operating at 1490 nm and showing 3 dBm launch power to mimic a 2.5 Gb/s G-PON downstream signal. It is worth noting that this emitter does not include any spectral filter. The G-PON and DFOS (operating at 1550 nm) signals are mixed thanks to a Wavelength Division Multiplexing (WDM) filter (WDM1 in Fig. 1.b) before propagating in a 20.8 km long Standard Single-Mode Fiber (SSMF). A Variable Optical Attenuator (VOA) called VOA1 can tune the launch power of the DFOS. The signals are attenuated with a VOA called VOA2 before reaching the G-PON receiver (ONU Rx in Fig. 1.b), and the Bit Error Ratio (BER) can be extracted from the received GPON signal. A Clock Data Recovery (CDR) recovers the clock. A second WDM filter (WDM2 in Fig. 1.b) can be used to separate the G-PON from the DFOS.

### Results and discussions

During the first part of the experiments, the G-PON ONU Rx receives both the G-PON and the DFOS signals, considering an ONU without optical filter (WDM2). Fig. 2.a shows the BER vs. the G-PON Received Optical Power (ROP) results for different DAS launch powers, for a 500 ns pulse width repeated every 250  $\mu$ s. The launch power

on Fig. 2.a is deliberately low (below 0 dBm) to prevent G-PON Rx damages, and not compatible with practical DAS measurements. In that configuration, when the DAS launch power gets above -6 dBm, a significant impact is observed on the BER measurements: the DAS blinds the G-PON receiver. This threshold appears at higher values when the pulse width decreases, as shows Fig. 2.b: for a fixed ROP (-30.2 dBm), the DAS pulse blinds the receiver for a lower duration, thus limiting its impact on the BER. The pulse width of the DAS to achieve 1-100 meter resolution is typically in the  $t_p = 10$ -1000 ns range, with a peak launch power in the 15-30 dBm range, and the repetition rate is typically  $T = 250$   $\mu$ s for 25 km coverage to avoid backscattered pulses collision. For very high pulses which totally blind the receiver for the pulse duration, the BER during reaches  $5 \cdot 10^{-1}$  for the pulse duration, as the zeros are above the decision threshold. Then, the average BER can be expressed as  $BER = (T - t_p)/T \times 10^{-9} + t_p/T \times 5 \cdot 10^{-1} \approx 0.5 \times t_p/T$  (assuming the notch-free BER= $10^{-9}$ ). In our configuration, this average BER should then be about  $10^{-5}$  and  $10^{-3}$  for pulses width of 5 ns and 500 ns, respectively. Then, the results show that a proper wavelength isolation is necessary to allow both G-PON and DAS to operate simultaneously. Generally speaking, DAS systems can only be used with PON technologies (including XGS-PON and 50G-PON) when employing ONUs with integrated spectral band filters filtering out the DAS channel.

An optical filter is then inserted in front of the receiver to isolate the G-PON ("WDM2" on Fig. 1). Fig. 2.c and .d show the DAS pulse and the corresponding captured G-PON frame when the DAS launch power is set to 27.3 dBm: a significant notch can be observed in the G-PON data, which duration corresponds to the pulse duration. This notch originates from power depletion due to SRS, since both signals are spectrally isolated. The SRS is an inelastic phenomenon that transfers energy from a low wavelength signal (G-PON) to a higher wavelength signal (DAS). Varying the states of polarization of the launch signals did not affect



**Fig. 3:** a) PON wavelength plan above 1400 nm, and relative Raman gain coefficient. b) GPON depletion depending on the DAS power (20.8 km SSMF). c) BER vs. ROP for various DAS power ( $\lambda_{DAS}=1550$  nm).

the depth of the notch ; the polarization coverage was averaged during propagation.

If a DAS system was to be introduced in access networks, it would operate in C-band or L-band for commercial reasons. Indeed, the launch signal needs to be boosted and the backscattered signal needs amplification too, requiring low noise and affordable amplifiers, such as EDFAs. As a result, the SRS would be triggered accordingly. Fig. 3.a summarizes the ITU-T PON wavelength plan in the 1400-1750 nm region, along with the Raman gain spectra for  $\lambda_{DAS}=1550$  nm and  $\lambda_{DAS}=1595$  nm. It can be observed that the downstream G-PON will be the main victim of the DAS introduction, as its power will be depleted, the normalized Raman gain being of 45% of the maximum if  $\lambda_{DAS}=1550$  nm and 100% when  $\lambda_{DAS}=1595$  nm. Most of the PON technologies operate in O-band (upstream G-PON, XGS-PON and 50G-PON, and downstream 50G-PON), away from potential SRS interactions when the DAS operates in C/L-bands. The notch duration is similar to the pulse duration, since the refractive index difference of SSMF is about  $\Delta n \approx 3.10^{-4}/100$  nm (vendor dependent): after 20 km, the propagation difference of the DAS and downstream G-PON is  $\approx 15$  ns, which is low compared to the DAS pulse width on Fig. 2.d.

The depth of the notch was measured for various DAS launch powers; see Fig. 3.b. It goes as high as 5 dB for  $P_{DAS}=27.3$  dBm and matches the usual SRS model [10, 11] with an effective area of  $80 \mu m^2$  and a Raman gain of  $5.4 \cdot 10^{-17}$  m/mW. Simulations of Fig. 3.b also shows the notch depth if the DAS was operating in the L-band ( $\lambda_{DAS}=1595$  nm). Simulations also showed that the gain on the DAS remained below 0.03 dB (thus negligible) in the considered range, whether  $\lambda_{DAS}=1595$  nm or  $\lambda_{DAS}=1550$  nm. As a result, a significant impact is observed on the BER, despite the wavelength isolation, see Fig. 3.c: for  $P_{DAS}$  launch powers higher than 20 dBm, the penalty on the GPON receiver sensitivity at BER=10<sup>-9</sup> exceeds 2 dB (see also in red on Fig. 3.b). When the notch is deep enough, the receiver's electronic (the Transimpedance Amplifier (TIA)) does not get enough time to adapt

its decision threshold: all the ones for the pulse duration are considered as zeros, and those errors are repeated at the pulse rate. The BER eventually reaches  $\approx 0.5 \times t_p/T$  for deep notches, as previously described. It is worth noting that the BER=10<sup>-9</sup> threshold is the typical sensitivity BER reference, as G-PON usually does not require the Forward Error Correction (FEC) to be enabled. Enabling the FEC during DAS measurements could be seducing since its threshold in G-PON is 10<sup>-4</sup> (RS[255,239]). However, the previously shown notches assume flashes of co-located errors (several tenths of nanoseconds or more, depending on the pulse width) and not isolated errors spread along the frame, such as in a usual noisy channel, with which FECs cannot deal.

### Conclusions

We studied the possibility to use DFOS systems based on high power backscattered pulses such as DAS along with PON. Wavelength isolation (i.e. a filter at the receiver side of the ONU) is generally embedded in PONs to allow the coexistence of different PON generations. Here, we showed that introducing DAS would make mandatory to filter out the DAS pulse. Also, considering the current commercial implementations, C-band DASs are preferred to L-bands to mitigate the SRS impact. However, high-power DAS pulses introduce significant notches in PONs, mainly in downstream G-PON, through the process of SRS. Mitigating the notches depth is possible in reducing the launch pulse peak power below 20 dBm, even more considering the limited PONs reach (20 km max) which might not require the full DAS power. Reducing the DAS pulse's width and repetition rate at the cost of signal to noise ratio can also limit the impact. Also, recent advances showed that exploiting wavelength and code orthogonality lowers the non-linear interaction between co-propagating sensing and data signals [12]. In conclusion, we showed that DAS techniques must be properly adapted to PON networks and then exploit the huge quantity of already deployed fiber access networks, not only in using "dark" fibers, but also exploiting active FTTH fibers.

## Acknowledgements

This work is part of the Lab'Optic, a joint research initiative between Orange Research and the French UMRs 6285 Lab-STICC and 6082 Foton Institute (IMT Atlantique, ENIB and ENSSAT).

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